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title: "Riemann Hypothesis PI-Cycle Link: Zeta Zero Mapping to UQFF Buoyancy Oscillations with PImath Encryption"

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Riemann Hypothesis PI-Cycle Link

Abstract

We establish a mapping between the non-trivial zeros of the Riemann zeta function $\zeta(\frac{1}{2} + it_k) = 0$ and UQFF buoyancy oscillation cycles. The buoyancy field evaluated at each zero:

$$F_{U,Bi,i}(t_k) = \sum_{n=1}^N B_n \sin(t_k \ln n)$$

where $B_n = B_0/n^2$ are buoyancy Fourier coefficients, defines a PI-cycle period $T_\pi(k) = 2\pi/t_k$. We introduce a PImath encryption layer using SHA-256 hash chains anchored to π -digit positions, producing tamper-evident verification of each physics computation. The fundamental buoyancy period $T_\pi(t_1) = 2\pi/14.1347 \approx 0.4446$ s defines the Riemann sector oscillation timescale.

1. Introduction

The Riemann hypothesis — that all non-trivial zeros of $\zeta(s)$ satisfy $\Re(s) = \frac{1}{2}$ — has deep connections to the distribution of prime numbers and, through the explicit formula, to quantum chaos. The UQFF framework provides a physical interpretation: each zeta zero corresponds to a resonant buoyancy mode in the 26-dimensional compressed gravity field.

2. Zeta Zero to Buoyancy Mapping

2.1 Buoyancy Fourier Expansion

The unified buoyancy field $F_{U,Bi,i}$ admits a Fourier decomposition over logarithmic frequencies:

$$F_{U,Bi,i}(t) = \sum_{n=1}^N B_n \sin(t \ln n)$$

The coefficients $B_n = B_0/n^2$ ensure L^2 convergence and reflect the $1/n^2$ decay of gravitational buoyancy contributions from successive harmonics.

2.2 Evaluation at Zeta Zeros

At each non-trivial zero t_k (where $\zeta(\frac{1}{2} + it_k) = 0$), the buoyancy field takes specific values. The first 20 zeros yield:

k	t_k	$F_{U,Bi,i}(t_k)$	$T_\pi(k)$
1	14.1347	computed	0.4446 s
2	21.0220	computed	0.2989 s
3	25.0109	computed	0.2513 s
...	...	...	...

2.3 PI-Cycle Period

The fundamental oscillation period in the Riemann sector:

$$T_\pi(k) = \frac{2\pi}{t_k}$$

The gap between consecutive periods encodes prime number distribution information via the explicit formula:

$$\psi(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \ln(2\pi) - \frac{1}{2} \ln(1 - x^{-2})$$

3. PImath Encryption Layer

3.1 Protocol

Each buoyancy computation is encrypted using a π -anchored SHA-256 scheme:

1. Extract 64-character segment from π digits starting at position $p_k = (k \cdot 7) \bmod (L - 64)$
2. Encode $F_{U,Bi,i}(t_k)$ as UTF-8 byte string
3. XOR the π -segment bytes with the F_U bytes
4. Compute $H_k = \text{SHA-256}(\pi[p_k : p_k + 64] \oplus F_U \text{ bytes})$

3.2 Hash Chain

The hash chain $\{H_1, H_2, \dots, H_K\}$ provides: - **Tamper evidence**: modifying any computation invalidates all subsequent hashes - **π -anchoring**: binding computations to irrational number positions prevents forgery - **Verifiability**: any party with π digits and the algorithm can independently verify

3.3 Cryptographic Properties

The use of SHA-256 provides 128-bit collision resistance. The π -digit anchoring adds computational binding — to forge a hash, an adversary must find a SHA-256 pre-image that simultaneously satisfies the π -digit XOR constraint.

4. Buoyancy-Zero Spectral Correspondence

The spectral density of buoyancy modes:

$$\rho_B(\omega) = \sum_k \delta(\omega - t_k) |F_{U,Bi,i}(t_k)|^2$$

follows the GUE (Gaussian Unitary Ensemble) pair correlation predicted by Montgomery's conjecture, providing a physical realisation of random matrix theory in the buoyancy spectrum.

5. Implications

The Riemann PI-cycle link suggests: 1. The distribution of zeta zeros encodes the structure of UQFF buoyancy resonances 2. The Riemann hypothesis, if true, implies a specific symmetry in the buoyancy spectrum 3. PImath encryption creates a verifiable record of all buoyancy computations anchored to π

References

- PAPER_971: Yang-Mills Mass Gap UQFF Derivation
- PAPER_1108: VDS/DVP/BH Unified Number System
- Montgomery, H.L. (1973). The pair correlation of zeros of the zeta function. Proc. Symp. Pure Math. 24
- Riemann, B. (1859). Über die Anzahl der Primzahlen unter einer gegebenen Grösse